

Here is the refined architecture for a pure **Productivity & Research Augmentation** tool.

1. The Core Philosophy: "State, Not Signal"

Your goal is to reduce the **"Time to Insight."**

- **Old Way (Manual):** PM asks a question → Analyst spends 4 hours pulling data, aligning dates, and building a chart → PM sees the chart.
- **New Way (Your Tool):** PM asks a question → Agent spends 30 seconds running the Python scripts → PM sees the chart + a synthesized summary of the *facts*.

2. The Refined Architecture: The "Digital Analyst Team"

We keep the **Vertical Desks**, but we change their mandate. They are no longer "predicting"; they are **measuring**.

Desk 1: The Sovereign Rates Analyst (The "Curve" Specialist)

- **Mandate:** "Tell me the shape of the money."
- **The "Productivity" Automation:**
 - **Manual Task:** Bootstrapping a zero-coupon curve from raw bond prices and calculating the 2s10s30s fly.
 - **Agent Action:** Instantly calls `curve_builder.py`.
- **Output:** "US 2s10s Spread is -40bps (Inverted). The curve has steepened 10bps this week, driven entirely by the front end rallying." (Fact, not advice).

Desk 2: The FX & Commodities Analyst (The "Flow" Specialist)

- **Mandate:** "Tell me the cost of positioning."
- **The "Productivity" Automation:**
 - **Manual Task:** Checking the "Carry" (interest differential) vs. the "Vol" (cost of protection) for 10 currency pairs.
 - **Agent Action:** Instantly calls `carry_screener.py`.
- **Output:** "USD/JPY offers 5% annualized carry. However, 1-month implied volatility is in the 90th percentile. The 'Carry-to-Risk' ratio is currently 0.4 (Low)."

Desk 3: The Volatility Analyst (The "Structure" Specialist)

- **Mandate:** "Tell me what the market fears."
- **The "Productivity" Automation:**
 - **Manual Task:** Pulling option chains to see if Puts are more expensive than Calls (Skew).
 - **Agent Action:** Instantly calls `skew_calculator.py`.
- **Output:** "Skew has flipped positive. The market is now paying a premium for Calls (Upside) rather than Puts (Downside). This is a 3-sigma shift from last month."

Desk 4: The Macro/Central Bank Analyst (The "Context" Specialist)

- **Mandate:** "Tell me what the policymakers are saying."
- **The "Productivity" Automation:**

- **Manual Task:** Reading 3 different speeches to find if they mentioned "Neutral Rate."
- **Agent Action:** Instantly calls nlp_search.py.
- **Output:** "In the last 48 hours, 3 Fed officials mentioned 'Neutral Rate'. All 3 suggested it might be higher than previously thought."

3. The New "Orchestrator": The Synthesis Engine

The Orchestrator does not recommend a trade. Instead, it performs **Information Fusion**. It arranges the data from the desks into a **coherent dashboard** for the human PM.

The Scenario: "The Carry Trade Check"

- **PM Query:** "Give me a deep dive on the USD/JPY Carry Trade setup."
- **The Agent's Workflow:**
 1. **Orchestrator:** Asks Rates Desk for yield spreads, FX Desk for spot momentum, Vol Desk for crash risk, and Macro Desk for BOJ policy tone.
 2. **Synthesis (The Output):**
 - ♦ **Yields:** "Spread is +350bps (Attractive)."
 - ♦ **Positioning:** "CFTC Data shows net longs are at a 5-year high (Crowded)."
 - ♦ **Risk:** "Vol Skew implies the market is pricing a significant risk of a rapid Yen strengthening."
 - ♦ **Policy:** "BOJ Governor Ueda's speech yesterday was scored as 'Hawkish'."
 3. **The "Productivity" Kick:** It generates a single PDF/Dashboard with:
 - ♦ A chart of the Yield Spread vs. Spot.
 - ♦ A chart of the Implied Volatility Surface.
 - ♦ A bulleted summary of the "Bearish" vs. "Bullish" data points.

Value Proposition: "I just saved you 3 hours of Excel work and data gathering. You (The Human) can now make the decision."

4. Why This Pitch Wins with Dr. Karri

This framing is actually **stronger** for a researcher because it focuses on **Data Integrity** and **Signal Processing** rather than "AI Magic."

The Pitch:

"I am building a '**Computational Research Stack**'.

In Macro Trading, analysts waste 80% of their time on **Data Wrangling** (finding tickers, aligning dates, cleaning noise).

My system uses Vertical Agents to automate this wrangling.

- The **Rates Agent** owns the Yield Curve models.
- The **Vol Agent** owns the Surface models.

The Research Problem I need your help with is the **Synthesis Layer**: When the Vol

Agent says 'Risk High' and the Macro Agent says 'Risk Low', how do we mathematically present this divergence to the user without hallucinating a resolution? I want to use **Submodular Optimization** to select the most relevant *evidence* for the user to see, rather than overwhelming them."

CONTEXT FOR NEW SESSION: The Macro Analyst Copilot Project

1. Project Overview & Mission I am building a "**Digital Macro Analyst Team**"—a multi-agent AI system designed to augment the workflows of discretionary global macro traders.

- **The Problem:** Traders and analysts spend 80% of their time on "grunt work"—finding Bloomberg tickers, aligning time-series dates, cleaning data, and maintaining Excel models—leaving only 20% for actual strategic thinking.
- **The Solution:** A productivity tool that acts as a "Text-to-Analysis" engine. It translates natural language queries into executable Python code to fetch live data, run deterministic financial models, and synthesize the results.
- **Key Constraint:** The tool does **NOT** make trade recommendations (e.g., "Buy Apple"). It focuses purely on **State Measurement** and **Dislocation Detection** (e.g., "Yields are 2 standard deviations away from fair value").

2. The Architecture: "The Vertical Desk Model" The system mimics the structure of a hedge fund trading floor. It consists of 4 specialized "Vertical Agents" (Desks) and 1 "Orchestrator."

- **Definition of an Agent:** Each agent is an encapsulated microservice containing:
 1. **Domain Knowledge** (LLM Context).
 2. **Private Data Layer** (Specific Bloomberg Tickers/Databases).
 3. **Deterministic Toolset** (Python scripts for math/modeling).

3. The 4 Vertical Pillars (The Experts)

- **Agent A: The Sovereign Rates Desk (The Curve Specialist)**
 - **Mandate:** Monitor government bond yields, OIS curves, and swap spreads.
 - **Python Toolkit:** `curve_builder.py` (Bootstrapping zero-coupon curves), `pca_model.py` (Decomposing moves into Level, Slope, Curvature).
 - **Output:** "US 2s10s spread has steepened 15bps; driven by front-end rally."
- **Agent B: The FX & Commodities Desk (The Flow Specialist)**
 - **Mandate:** Monitor currency pairs, commodity futures, and cross-border flows.

- **Python Toolkit:** carry_screener.py (Calculating implied yield differentials), z_score_scanner.py (Spotting mean-reversion setups).
- **Output:** "Long USD/JPY Carry is +5% annualized, but Spot is 2 Sigma overbought."
- **Agent C: The Volatility & Structure Desk (The Risk Specialist)**
 - **Mandate:** Monitor option chains, skew, gamma, and implied volatility surfaces.
 - **Python Toolkit:** vol_surface.py (SABR model fitting), skew_tracker.py (Call vs. Put premiums).
 - **Output:** "Skew has flipped positive; market is bidding up upside calls over downside puts."
- **Agent D: The Central Bank & Macro Desk (The Policy Specialist)**
 - **Mandate:** Monitor policy intent (Speeches/Minutes) and economic data (CPI/GDP).
 - **Python Toolkit:** nlp_scorer.py (Hawk/Dove sentiment scoring), nowcast_model.py (Real-time GDP estimation).
 - **Output:** "Fed Chair Powell's speech scored 8/10 Hawkish, diverging from market pricing."
- 4. The Orchestration Layer (The "Reasoning Agent")**
 - **Role:** Acts as the "Digital Portfolio Manager." It does not touch raw data.
 - **Function:** It parses the user's high-level intent, delegates tasks to the relevant Vertical Agents, and synthesizes their outputs into a coherent dashboard.
 - **Synthesis Logic:** It looks for **Divergences**.
 - *Example:* "The Macro Desk says Inflation is rising, but the Rates Desk says Breakevens are falling. **Conclusion:** Market is complacent."
- 5. Technical Stack & Data Flow**
 - **Input:** Natural Language Queries.
 - **Data Source:** Bloomberg Terminal API (blpapi) + News Feeds.
 - **Execution:** Python (Pandas, SciPy, QuantLib) for all math. The LLM writes the scaffolding code to call these libraries.
 - **Output:** Streamlit Dashboard / PDF Briefing containing deterministic charts, data tables, and bulleted text summaries

First step:

To summarize the first phase of the **Macro Analyst Copilot Project**, we need to frame it through the lens of a **"Suitcase Architecture."** This approach ensures we maintain institutional-grade data integrity while bypassing the physical limitations of the Bloomberg Terminal.

Here is the high-level summary of the **Data Ingestion Layer**:

1. The Architecture: "The Suitcase Model"

Since the Bloomberg Terminal is a "landlocked" asset (tied to a specific physical machine), we are building a decoupled ingestion system.

- **The Library (Ingestion):** A script runs on the Terminal to fetch "snapshots" of historical and live data.
- **The SSD (Storage):** Data is stored in a local, portable **SQLite** database. This provides an air-gapped, ACID-compliant data store that we can move between the library and our development environments.
- **The Home (Analysis):** Our agent logic (Rates, FX, Vol, Macro) reads from this local DB, allowing us to build complex models without needing a 24/7 Bloomberg connection.

2. The Universal Bloomberg Client

We are building a **Data-Agnostic Wrapper** around the Bloomberg API (**blpapi**). This client isn't hardcoded for specific tickers; it's a gateway that translates our internal requests into Bloomberg-specific queries.

- **Capabilities:** It will handle **BDH (Bulk Data History)** for time-series and **BDP (Bulk Data Point)** for reference/static data.
- **Normalization:** The client cleans the raw Bloomberg output—handling market holidays, scale differences (e.g., 4.5% vs 0.045), and missing data—before it hits the DB.

3. Initial Focus: G4 Sovereign Rates

For the first build, we are populating the DB with the **G4 Sovereign Rates** universe (US, Germany, UK, Japan).

- **The Knots:** 2Y, 5Y, 10Y, and 30Y yields.
- **The Fields:** We aren't just pulling price; we are pulling **Yield to Maturity (YLD_YTM_MID)** and **Modified Duration (DUR_ADJ_MID)**. This allows Agent A to perform PCA (Principal Component Analysis) to decompose curve moves into Level, Slope, and Curvature.

4. The Tech Stack

- **Language:** Python.
- **Database:** SQLite (chosen for its zero-latency local performance and 140TB scalability).
- **Interface:** blpapi (Bloomberg Desktop API).
- **Schema:** A 3-table structure:
 1. ticker_reference: Metadata about the assets.
 2. market_data: The historical time-series.
 3. news_stories: Pre-structured for Agent D's NLP ingestion of speeches/minutes.

